Algebra Quadratics Cheat Sheet

Unit Focus: *graphing and solving quadratic functions*

1 KEY VOCABULARY

quadratic	A polynomial function of degree 2. In the form $f(x) = ax^2 + bx + c$ where $a \neq 0$.
parabola	The U-shaped graph of a quadratic function.
vertex	The highest or lowest point on the parabola (h, k).
axis of symmetry	A vertical line that goes through the vertex. Equation: $x = h$.
x-intercepts (zeros, roots)	The points where the graph crosses the x-axis. Occur when $f(x) = 0$.
y-intercept	The point where the graph crosses the y-axis. Occurs at $(0, c)$.
maximum	The highest point of a downward-opening parabola (when $a < 0$).
minimum	The lowest point of an upward-opening parabola (when $a > 0$).

2 RECOGNIZING A QUADRATIC FUNCTION

A quadratic function is a function of the form

$$f(x) = ax^2 + bx + c \quad (a \neq 0)$$

- The highest power of the variable is 2.
- It graphs as a **parabola** (U-shaped).
- It can be written in standard form, factored form, or vertex form.

Examples:

- ✅ $f(x) = 3x^2 - 5x + 1$ (quadratic)
- ✅ $y = -2x^2$ (quadratic)
- ✅ $g(x) = (x - 4)(x + 1)$ (quadratic)
- ✅ $h(x) = (x + 3)^2 - 2$ (quadratic)
- ❌ $f(x) = 4x + 7$ (not quadratic)
- ❌ $y = x^3 - 1$ (not quadratic)
- ❌ $f(x) = \frac{2}{x} + 5$ (not quadratic)

A quadratic function always has an x^2 term and graphs as a parabola!

3 FORMS OF A QUADRATIC FUNCTION

Standard Form

$f(x) = ax^2 + bx + c$

- Easy to identify a, b, c
- Used to find intercepts, axis of symmetry, and vertex.

Example:

$f(x) = 2x^2 - 4x - 6$

$a = 2,\ b = -4,\ c = -6$

Factored Form

$f(x) = a(x - r_1)(x - r_2)$

- Shows the x-intercepts directly: $x = r_1$ and $x = r_2$.

Example:

$f(x) = (x - 3)(x + 1)$

x-intercepts: $x = 3$ and $x = -1$

Vertex Form

$f(x) = a(x - h)^2 + k$

- Shows the vertex directly: (h, k).
- Easy to graph.

Example:

$f(x) = -2(x + 4)^2 + 5$

Vertex: $(-4, 5)$

Axis of symmetry: $x = -4$

4 KEY FEATURES OF A PARABOLA

Example: Graph $f(x) = x^2 - 4x - 5$.

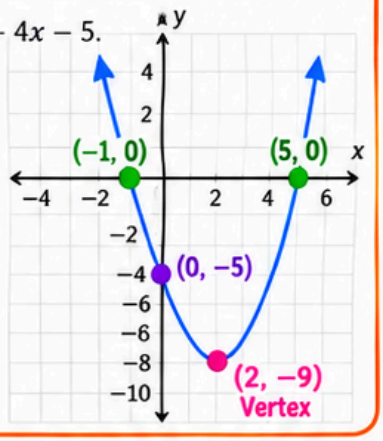
Opens: Up ($a = 1 > 0$)

Vertex: $(2, -9)$ (minimum)

Axis of symmetry: $x = 2$

x-intercepts: $(-1, 0)$ and $(5, 0)$

y-intercept: $(0, -5)$



5 FINDING THE AXIS OF SYMMETRY AND VERTEX

For $f(x) = ax^2 + bx + c$:

Axis of symmetry: $x = -\frac{b}{2a}$

Vertex: (h, k) where $h = -\frac{b}{2a}$ and $k = f(h)$.

Example: Find the axis of symmetry and vertex of $f(x) = 2x^2 + 8x - 6$.

Step 1: $x = -\frac{b}{2a} = -\frac{8}{2(2)} = -\frac{8}{4} = -2$

Step 2: Find $f(-2)$: $2(-2)^2 + 8(-2) - 6$
$= 2(4) - 16 - 6 = 8 - 16 - 6 = -14$

Step 3: Vertex: $(-2, -14)$

Axis of symmetry: $x = -2$

Vertex: $(-2, -14)$

6 FINDING X-INTERCEPTS AND Y-INTERCEPT

x-Intercepts (Solve $f(x) = 0$)

Example: Find the x-intercepts of $f(x) = x^2 - 4x - 5$.

Step 1: Factor:
$x^2 - 4x - 5 = (x - 5)(x + 1)$

Step 2: Set each factor equal to 0:
$x - 5 = 0 \Rightarrow x = 5$
$x + 1 = 0 \Rightarrow x = -1$

x-intercepts: $(-1, 0)$ and $(5, 0)$

y-Intercept (Set $x = 0$)

Example: Find the y-intercept of $f(x) = x^2 - 4x - 5$.

Step 1: Set $x = 0$:
$f(0) = 0^2 - 4(0) - 5$

Step 2: Simplify:
$f(0) = -5$

y-intercept: $(0, -5)$

7 QUICK REMINDERS & COMMON MISTAKES

✅ DO REMEMBER:

- A quadratic has an x^2 term (degree 2).
- Use $x = -\frac{b}{2a}$ to find the axis of symmetry.
- The vertex is a maximum if $a < 0$ and a minimum if $a > 0$.
- Set $f(x) = 0$ to find x-intercepts (solve the equation).
- The y-intercept is $f(0) = c$.
- Check your answers on a graph or by substitution.

❌ COMMON MISTAKES:

- Forgetting that $a \neq 0$ (otherwise it's not quadratic).
- Using $x = \frac{b}{2a}$ instead of $x = -\frac{b}{2a}$.
- Confusing the vertex (h, k) with the axis of symmetry $x = h$.
- Thinking a parabola with $a > 0$ has a maximum (it has a minimum).
- Forgetting to set $f(x) = 0$ when finding x-intercepts.
- Arithmetic errors when factoring or substituting.

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Algebra Quadratics Cheat Sheet

More Practice: *solving quadratic equations*

1 KEY VOCABULARY

quadratic equation	An equation that can be written in the form $ax^2 + bx + c = 0$, where $a \neq 0$.
quadratic function	A function in the form $f(x) = ax^2 + bx + c$. Its graph is a parabola.
root (solution)	A value of x that makes the equation true (also called a zero or x-intercept).
vertex	The highest or lowest point on the parabola.
axis of symmetry	A vertical line through the vertex: $x = -\frac{b}{2a}$.
discriminant	The value $b^2 - 4ac$, which tells the number and type of real solutions.

2 SOLVING BY FACTORING

Factor the quadratic expression, set each factor equal to zero, and solve for the variable.

Example: Solve $x^2 - 5x - 6 = 0$.

Step 1: Factor: $(x - 6)(x + 1) = 0$

Step 2: Set each factor equal to zero: $x - 6 = 0$ or $x + 1 = 0$

Step 3: Solve: $x = 6$ or $x = -1$

If the quadratic factors easily, this is often the fastest method!

Answer: $x = 6, -1$

3 SOLVING BY SQUARE ROOTS

Use this method when the quadratic can be written in the form $(x - h)^2 = k$ or $x^2 = k$.

Example: Solve $x^2 - 16 = 0$.

Step 1: Isolate x^2: $x^2 = 16$

Step 2: Take square roots: $x = \pm\sqrt{16}$

Step 3: Simplify: $x = \pm 4$

$x = 4$ or $x = -4$

Remember: If $x^2 = k$, then $x = \pm\sqrt{k}$ (when $k \geq 0$).

Answer: $x = 4, -4$

4 QUADRATIC FORMULA

Use the quadratic formula to solve any quadratic equation of the form $ax^2 + bx + c = 0$.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Example: Solve $2x^2 + 4x - 6 = 0$.

Step 1: Identify a, b, c: $a = 2$, $b = 4$, $c = -6$

Step 2: Substitute: $x = \frac{-4 \pm \sqrt{4^2 - 4(2)(-6)}}{2(2)}$

Step 3: Simplify: $x = \frac{-4 \pm \sqrt{64}}{4} = \frac{-4 \pm 8}{4}$

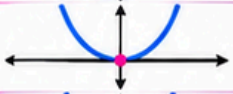
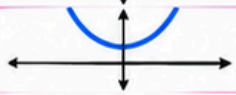
$x = 1$ or $x = -3$

Use this method when factoring is not easy!

Answer: $x = 1, -3$

5 THE DISCRIMINANT

The discriminant is the expression inside the square root of the quadratic formula: $D = b^2 - 4ac$. It tells you the number and type of real solutions.

Value of D	Number of Real Solutions	Graph
$D > 0$	Two distinct real solutions (two x-intercepts)	
$D = 0$	One real solution (double root, touches x-axis)	
$D < 0$	No real solutions (no x-intercepts)	

6 MATCHING FORMS TO INFORMATION

Different forms of a quadratic equation give different information.

Form	Example	Useful Information
Standard Form $ax^2 + bx + c$	$2x^2 - 5x - 3$	• Use with quadratic formula • Easy to find a, b, c • Use discriminant $(b^2 - 4ac)$
Factored Form $a(x - r_1)(x - r_2)$	$3(x + 2)(x - 5)$	• Solutions are r_1 and r_2 • x-intercepts: $x = -2, 5$ • Easy to graph
Vertex Form $a(x - h)^2 + k$	$4(x - 1)^2 - 9$	• Vertex is (h, k) • Axis of symmetry: $x = h$ • Shows minimum or maximum

7 SPECIAL TIPS & QUICK REMINDERS

- Always simplify before solving.
- Check for a common factor first.
- Make sure your solution actually works (substitute it back in).
- Solutions can be written as $x = a, b$ or $x = a, b, \ldots$
- Watch for extraneous solutions if you square both sides.
- For graphs, x-intercepts are the solutions (where $y = 0$).
- The axis of symmetry is $x = -\frac{b}{2a}$.
- The vertex gives the maximum or minimum value.

8 REAL-WORLD EXAMPLE

Quadratic equations model many real situations, like projectile motion, area problems, and profit.

Example: A ball is thrown upward from a height of 5 feet with the model $h(t) = -16t^2 + 48t + 5$. When does the ball hit the ground?

Step 1: Set $h(t) = 0$: $-16t^2 + 48t + 5 = 0$

Step 2: Solve (quadratic formula): $t = \frac{-48 \pm \sqrt{48^2 - 4(-16)(5)}}{2(-16)}$

Step 3: Simplify: $t \approx 3.1$ or $t \approx -0.1$

Step 4: Interpret: Time cannot be negative, so $t \approx 3.1$ seconds.

Answer: $t \approx 3.1$ s

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